MATH 205 LINEAR ALGEBRA
ASSIGNMENT 2

Q. no. 1

(a) Under what conditions
AB = BA, where

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

(b) If A is a matrix then

$A^m A^s = A^{m+s}$ for m, s positive integers.

Is this result true for negative integers also?

Justify your answer.

(c) If

$$\begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$$

and

$$\begin{bmatrix} 2 & 5 \\ 3 & 4 \end{bmatrix}$$

then $AB = AC = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$

but B ≠ C. Why?

Q. no. 2

Using the technique of forming a block matrix [A | I] and performing E.R.O.s such that

$[A \mid I]$ E.R.O.s $\rightarrow$ $[I \mid A^{-1}]$

find the inverse of the following where A is given by

(a)
$$\begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{bmatrix}$$

(b)
$$\begin{bmatrix} 1 & 2 & -4 \\ -1 & -1 & 5 \\ 2 & 7 & -3 \end{bmatrix}$$

Q. no. 3

Solve the following system of equations by reducing them to echelon form (Gaussian elimination method):

$$x + y + 2z = 9$$
$$2x + 4y - 3z = 1$$
$$3x + 6y - 5z = 0$$

Q. no. 4

Solve the following system by Gauss-Jordan elimination (reduced row-echelon form):

$$2x_1 + 2x_2 + 2x_3 = 0$$
$$-2x_1 + 5x_2 + 2x_3 = 1$$
$$8x_1 + x_2 + 4x_3 = -1$$

Q. no. 5

Reduce to

$$\begin{bmatrix} 2 & 1 & 3 \\ 0 & -2 & 7 \\ 3 & 4 & 5 \end{bmatrix}$$

reduced row echelon form without introducing any fractions.

Q. no. 6

Find two different row-echelon forms of

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$$